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Inventor(s)	Cheng; Tien-Ni

Motion simulating apparatus and actuating system

Abstract

A motion simulating apparatus and actuating system includes a base; a movable platform arranged above the base; a first cross arm assembly disposed on a first side between the base and the movable platform. The first cross arm assembly includes a first arm and a second arm that are cross-connected to each other; a second cross arm assembly disposed on a second side between the base and the movable platform relative to the first side. The second cross arm assembly includes a third arm and a fourth arm forming an X-shaped connection; an actuator connected obliquely to an upper part of the first arm and the base or the lower part of the third arm to drive the movable platform to move along a first direction or its opposite direction; and a rod is horizontally connected with the upper part of the first arm and an upper part of the third arm. The base, the movable platform, the first arm and the third arm form a substantially parallelogram when the actuator is actuating.

Inventors:	Cheng; Tien-Ni (Kaohsiung, TW)
Applicant:	Brogent Technologies Inc. (Kaohsiung, TW)
Family ID:	1000008820125
Assignee:	Brogent Technologies Inc. (Kaohsiung, TW)
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Primary Examiner: Garlen; Alexander K

Attorney, Agent or Firm: Birch, Stewart, Kolasch & Birch, LLP

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION(S)

(1) This application claims priority to Taiwan patent application no. 111120259 filed on May 31, 2022.

BACKGROUND

1. Field of the Invention

(2) The present invention relates to a motion simulating apparatus, and more particularly, to a motion simulating apparatus equipped with actuating systems.

2. Description of the Related Art

(3) Motion simulators may be typically used as amusement or training equipment. Generally, a motion simulator includes a plurality of actuators that are respectively disposed at different locations and along different axes of movements so as to be able to simulate motions along different axes. In general, a conventional actuator is installed vertically and has a large volume, such that the actuator requires a relatively tall space, which is disadvantageous to the scenario of limited space. Unfortunately, the conventional motion simulators are usually complex in construction and have relatively high manufacturing costs.

(4) Therefore, there is a need for a new construction that can solve the aforementioned issues.

SUMMARY

(5) The present application describes a motion simulating apparatus that can address at least the foregoing issues, and an actuating system incorporating the motion simulating apparatus.

(6) According to one aspect, a motion simulating apparatus described herein includes a base, a movable platform arranged above the base, a first cross arm assembly, a second cross arm assembly, an actuator and a rod. The first cross arm assembly is disposed on a first side between the base and the movable platform, wherein the first cross arm assembly includes a first arm and a second arm that are cross-connected to each other. The second cross arm assembly is disposed on a second side between the base and the movable platform relative to the first side, wherein the second cross arm assembly includes a third arm and a fourth arm that are cross-connected to each other. The actuator is connected obliquely to an upper part of the first arm and the base or a lower part of the third arm, configured to drive the movable platform to move along a first direction or an opposite direction of the first direction. The rod is horizontally connected with the upper part of the first arm and an upper part of the third arm, wherein the base, the movable platform, the first arm and the third arm substantially form a parallelogram when the actuator is actuating.

(7) According to another aspect, an actuating system described herein includes an occupant platform adapted to carry one or more occupants, and a plurality of the motion simulating apparatus. Wherein the movable platform of the motion simulating apparatus is pivotally connected to the occupant platform through a plurality of joint assemblies.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a front view illustrating an embodiment of a motion simulating apparatus;
- (2) FIG. 2 is a rear view illustrating an embodiment of a motion simulating apparatus;
- (3) FIG. 3 is a perspective view illustrating an embodiment of a motion simulating apparatus;
- (4) FIG. 4 is a perspective view illustrating an embodiment of a motion simulating apparatus;
- (5) FIG. 5 is a side view illustrating an embodiment of a motion simulating apparatus;
- (6) FIG. 6 is a perspective view illustrating an embodiment of a motion simulating apparatus;
- (7) FIG. 7 is a perspective view illustrating an embodiment of an actuating system; and
- (8) FIG. 8 is an enlarged view illustrating a portion of the actuating system.

DETAILED DESCRIPTION OF THE EMBODIMENTS

(9) FIG. 1 is a front view illustrating an embodiment of a motion simulating apparatus. FIG. 2 is a rear view illustrating an embodiment of a motion simulating apparatus. FIG. 3 and FIG. 4 are perspective views illustrating an embodiment of a motion simulating apparatus. FIG. 5 is a side view illustrating an embodiment of a motion simulating apparatus. FIG. 6 is a perspective view illustrating an embodiment of a motion simulating apparatus. FIG. 7 is a perspective view illustrating an embodiment of an actuating system. FIG. 8 is an enlarged view illustrating a portion of the actuating system. Referring to FIGS. 1-7, a first direction D1, a second direction D2 and a third direction D3 can be substantially orthogonal to each other. The motion simulating apparatus **100** can carry occupants or passengers and generate motions in different directions, which is applicable to various applications such as simulating platforms or amusement equipment. Referring to FIGS. 1-5, the motion simulating apparatus **100** can include a base **102**, a movable platform **104**, a first cross arm assembly **106**, a second cross arm assembly **108**, an actuator **110** and a rod **112**.

(10) The base **102** is disposed on a horizontal plane (the plane formed by the second direction D2 and the third direction D3). The movable platform **104** is arranged above the base plate **102** and is movably connected to the base **102**. The first cross arm assembly **106** and the second cross arm assembly **108** are disposed between the base **102** and the movable platform **104**. The first cross arm

assembly **106** and the second cross arm assembly **108** are respectively arranged on two opposite sides of the base **102**. The first cross arm assembly **106** can include a first arm **114** and a second arm **116** that are cross-connected to each other. The second cross arm assembly **108** can include a third arm **118** and a fourth arm **120** that are cross-connected to each other. According to an embodiment of the present invention, the first arm **114** can be pivotally connected to the second arm **116** through a first pivot member **124**, and the third arm **118** can be pivotally connected to the fourth arm **120** through a second pivot member **126**, so that first cross arm assembly **106** and the second cross arm assembly **108** can cooperate with the actuation of the actuator **110**. According to an embodiment of the present invention, the first cross arm assembly **106** and the second cross arm assembly **108** can both form an X-shaped connection. More specifically, the first arm **114** and the second arm **116** can be connected in an X shape to form a scissor lift structure, and the third arm **118** and the fourth arm **120** can be connected in an X shape to form a scissor lift structure, so as to cooperate with the operation of the actuator **110** to be extended and retracted.

(11) According to an embodiment of the present invention, the first arm **114** can include a first sub-arm **114A** and a second sub-arm **114B** spaced apart along the third direction D3, and the first sub-arm **114A** and the second sub-arm **114B** can be pivotally connected through the first pivot member **124**. In addition, the second arm **116** can include a third sub-arm **116A** and a fourth sub-arm **116B** spaced apart along the third direction D3, and the third sub-arm **116A** and the fourth sub-arm **116B** can be pivotally connected through the first pivot member **124**. According to an embodiment of the present invention, the lower side of the first arm **114** is fixedly disposed on the base **102** at a first fixed end **130**. According to embodiment of the present invention, the first sub-arm **114A** and the second sub-arm **114B** can also be connected to each other through a first connecting member **122**, and the first connecting member **122** can connect the first sub-arm **114A** with the second sub-arm **114B** by a fastener (such as a linkage rod or a screw rod, but the present invention is not limited thereto).

(12) According to an embodiment of the present invention, the third arm **118** can include a fifth sub-arm **118A** and a sixth sub-arm **118B** spaced apart along the third direction D3, and the fifth sub-arm **118A** and the sixth sub-arm **118B** can be pivotally connected through the second pivot member **126**. In addition, the fourth arm **120** can include a seventh sub-arm **120A** and an eighth sub-arm **120B** spaced apart along the third direction D3, and the seventh sub-arm **120A** and the eighth sub-arm **120B** can be pivotally connected through the second pivot member **126**. According to an embodiment of the present invention, the lower side of the third arm **118** is fixedly disposed on the base **102** at a second fixed end **132**.

(13) Referring to FIGS. 3-4, according to an embodiment of the present invention, the first sub-arm **114A** and the third sub-arm **116A** can be cross-connected to each other, and the second sub-arm **114B** and the fourth sub-arm **116B** can be cross-connected to each other. According to an embodiment of the present invention, the fifth sub-arm **118A** and the seventh sub-arm **120A** can be cross-connected to each other, and the sixth sub-arm **118B** and the eighth sub-arm **120B** can be cross-connected to each other. According to an embodiment of the present invention, the first sub-arm **114A** and the third sub-arm **116A** forming the X-shaped connection can be rotated relative to each other through the first pivot member **124** (such as a shaft, but the present invention is not limited thereto). The second sub-arm **114B** and the fourth sub-arm **116B** forming the X-shaped connection can be rotated relative to each other through the first pivot member **124**. The fifth sub-arm **118A** and the seventh sub-arm **120A** forming the X-shaped connection can be rotated relative to each other through the second pivot member **128** (such as a shaft, but the present invention is not limited thereto). The sixth sub-arm **118B** and the eighth sub-arm **120B** forming the X-shaped connection can be rotated through the second pivot member **128**. As a result, the X-shaped connection is able to cooperate with the operation of the actuator **110**.

(14) Referring to FIGS. 1-5, the actuator **110** can be connected obliquely to an upper part of the first arm **114** and the base **102**. Alternatively, referring to FIG. 6, the actuator **110** can be connected

obliquely to the upper part of the first arm **114** and a lower part of the third arm **118**. Please refer to FIGS. **1-6**, the actuator **110** can drive the movable platform **104** to move along the first direction **D1** or the opposite direction of the first direction **D1**. More specifically, one side of the actuator **110** is connected with the first connecting member **122**, and another side of the actuator **110** is connected with the base **102** or the lower part of the third arm **118** (as shown in FIG. **6**). When the actuator **110** is extended or retracted, the actuator **110** can drive the movable platform **104** to move up and down relative to the base **102**, so as to achieve the heave motion. The actuator **110** is arranged as a reclining arrangement instead of a vertical or flat arrangement, which can effectively reduce the height of the motion simulating apparatus, such that the height of the installation space occupied by the motion simulating apparatus can also be reduced. The actuator **110** can be an electric cylinder, an oil cylinder, a pneumatic cylinder, a motor (such as geared motor or stepper motor) or other device capable of transforming energy into a controllable displacement.

(15) Referring to FIGS. **1-6**, the rod **112** can substantially connect the upper part of the first arm **114** and the upper part of the third arm **118** horizontally. When the actuator **110** is actuated, the base **102**, the movable platform **104**, the first arm **114** and the third arm **118** are substantially in the shape of a parallelogram. According to an embodiment of the present invention, the base **102**, the movable platform **104** and the rod **112** are all substantially parallel to the second direction **D2**, the length of the rod **112** is substantially equal to a distance **H** between the first fixed end **130** and the second fixed end **132**, and the first arm **114** and the third arm **118** are also substantially parallel to each other and have the same length. Therefore, the first arm **114**, the rod **112**, the third arm **118** and the two fixed ends (the first fixed end **130** and the second fixed end **132**) can form a parallelogram substantially. In this way, when the actuator **110** is extended or retracted, the movable platform **104** can be ensured to be moved up and down horizontally and stably. According to other embodiments, the configuration of the rod **112** may be one rod or multiple rods, and the purpose of the configuration of the rod(s) is to enable the first cross arm assembly **106** and the second cross arm assembly **108** to be move smoothly and synchronously, so as to ensure that the movable platform **104** can be moved up and down horizontally and stably.

(16) According to an embodiment of the present invention, a first roller assembly **134** comprising multiple rollers can be disposed on the upper part of the first arm **114** and the lower part of the second arm **116** respectively, and a second roller assembly **136** comprising multiple rollers can be disposed on the upper part of the third arm **118** and the lower part of the fourth arm **120** respectively. Referring to FIGS. **1, 3, 4** and **6**, according to an embodiment of the present invention, the first cross arm assembly **106** with the first roller assembly **134** is arranged on the left side of the actuator **110**, and the second cross arm assembly **108** with the second roller assembly **136** is arranged on the right side of the actuator **110**. The installation directions of the first cross arm assembly **106** and the second cross arm assembly **108** are the same direction. Referring to FIG. **1**, the first roller assembly **134** of the first cross arm assembly **106** is on the right side relative to the fixed end (such as the first fixed end **130**), and the second roller assembly **136** of the second cross arm assembly **108** is also on the right side relative to the fixed end (such as the second fixed end **132**). It is worth noting that the first cross arm assembly **106** and the second cross arm assembly **108** are not limited to the aforementioned arrangement. According to another embodiment, the first cross arm assembly **106** with the first roller assembly **134** and the second cross arm assembly **108** with the second roller assembly **136** may be on the left side relative to the fixed end, so that the side of the rod **112** connected to the first arm **114** and the third arm **118** is movable relative to the base **102**. The actuator **110** can drive the first arm **114** and the third arm **118** through the rod **112** to move along the second direction **D2** or the opposite direction of the second direction **D2**, in order to elevate the movable platform **104**. The rod **112** is configured to drive the first cross arm assembly **106** and the second cross arm assembly **108** to be extended and retracted stably in cooperation with the operation of the actuator **110**, so the rod **112** must be arranged on the side with the roller assembly (the first roller assembly **134** or the second roller assembly **136**), that is, the

arrangement of the upper part of the first arm **114** and the upper part of the third arm **118** as shown in FIGS. **1-6**.

(17) Please referring to FIG. **3** more specifically, first rollers **134a**, **134b** can be disposed on the upper part of the first sub-arm **114A** and the upper part of the second sub-arm **114B** respectively. The first rollers **134a**, **134b** can be installed on a rod between the upper part of the first sub-arm **114A** and the upper part of the second sub-arm **114B**. First rollers **134c**, **134d** can be disposed on the lower part of the third sub-arm **116A** and the lower part of the fourth sub-arm **116B** respectively. Referring to FIGS. **3-5**, second rollers **136a**, **136b** can be disposed on the upper part of the fifth sub-arm **118A** and the upper part of the sixth sub-arm **118B** respectively, and the second rollers **136a**, **136b** can be installed on a rod between the upper part of the fifth sub-arm **118A** and the upper part of the sixth sub-arm **118B**. Second rollers **136c**, **136d** can be disposed on the lower part of the seventh sub-arm **120A** and the lower part of the eighth sub-arm **120B** respectively so as to facilitate compact assembly. Furthermore, the arrangement of the first roller assembly **134** and the second roller assembly **136** can cooperate with the extension or retraction of the actuator **110**, the first roller assembly **134** and the second roller assembly **136** can perform horizontal reciprocating movement, so that the movable platform **104** can be moved up and down smoothly.

(18) Referring to FIGS. **1-6**, according to an embodiment of the present invention, a first supporting member **138** can be disposed between the second arm **116** and the base **102**, and a second supporting member **140** can be disposed between the fourth arm **120** and the base **102**. More specifically, the first supporting member **138** can be disposed on the base **102** and disposed under the first rollers **134c**, **134d** of the third sub-arm **116A** and the fourth sub-arm **116B**; and the second supporting member **140** can be disposed on the base **102** and disposed under the second rollers **136c**, **136d** of the seventh sub-arm **120A** and the eighth sub-arm **120B**, thereby providing support and moving space for the first rollers **134c**, **134d** and the second rollers **136c**, **136d**. It is worth noting that the two supporting members **138**, **140** are not limited to the aforementioned arrangement.

(19) In conjunction with FIGS. **1-6**, FIG. **7** shows a perspective view illustrating an embodiment of an actuating system. FIG. **8** shows an enlarged view illustrating a portion of the actuating system. Referring to FIGS. **1-7**, the actuating system **150** can include an occupant platform **160**, a plurality of motion simulating apparatus **100** and a plurality of joint assemblies **170**. The actuating system **150** can carry occupants or passengers and generate motions along different directions, which is applicable for applications such as simulating platforms or amusement equipment. The occupant platform **160** is pivotally connected to the plurality of motion simulating apparatus **100** via the plurality of joint assemblies **170**, and the occupant platform **160** is applied to carry one or more occupants or passengers. The occupant platform **160** is pivotally connected to the plurality of joint assemblies **170**, so that the occupant platform **160** can pivot around the second direction **D2** relative to the plurality of joint assemblies **170** and the plurality of motion simulating apparatus **100**. It is noted that, the joint assembly **170** is a kind of joint structure. The joint structure being used for pivoting is known to those skilled in the art, so details related to this are not explained further herein.

(20) Referring to FIG. **7**, according to a symmetrical arrangement of an embodiment, the actuating system **150** can include two motion simulating apparatuses **100**, the two motion simulating apparatuses **100** may be symmetrically disposed at two opposite sides of the occupant platform **160** respectively, and pivotally connected to the occupant platform **160**. According to the configuration of the two motion simulating apparatuses **100**, the actuator **110** may apply a force on the occupant platform **160** so as to generate a torque to drive the occupant platform **160** to be rotated around the second direction **D2**, and/or moved along the first direction **D1** or the opposite direction of the first direction **D1**. For example, the two actuators **110** can be operated to generate synchronous and identical procedure to cause the occupant platform **160** to be moved up and down along the first direction **D1** or the opposite direction of the first direction **D1** relative to the base **102**, and the two

actuators **110** can be operated to generate different procedures to cause the occupant platform **160** to rotate around a pivot axis of the second direction D2 relative to the base **102**. Based on the arrangement of the occupant platform **160**, the simulating apparatus **100** and the plurality of joint assemblies **170** as described herein, the actuating system **150** can have a simple construction capable of simulating heave and roll motions with the plurality of simulating apparatuses **100**. This configuration is applicable for simulating somatosensory experiences (such as elevator lifting, flying carpet, surfing, and flying, but are not limited thereto).

(21) Referring to FIGS. 7-8, according to an embodiment of the present invention, the actuating system **150** further includes a plurality of slide assemblies **180**, the slide assemblies **180** (such as slide rails but not limited thereto) are fixed below one side of the occupant platform **160**, and the joint assemblies **170** is slidably disposed on the slide assemblies **180**, so that the actuating system **150** can perform substantially horizontal movement. In this way, when the motion simulating apparatus **100** simulates the motions of heave and roll, the occupant platform **160** can be slid relative to at least one of the plurality of joint assemblies **170**, so as to prevent the actuation system **150** from fracture due to rigidity issues. The arrangement of the sliding assembly **180** is not limited to the above embodiments.

(22) Referring to FIGS. 7-8, according to an embodiment of the present invention, the actuating system **150** can further include a plurality of pillars **190** disposed between the occupant platform **160** and the base **102**. A plurality of cushioning members **192** can also be disposed between the plurality of pillars **190** and the occupant platform **160**, so that when an abnormal mechanical operation is occurred on the actuation system **150**, the cushioning members **192** can buffer at least one impact between the occupant platform **160** and the motion simulating apparatus **100**. In this configuration, the range of the occupant platform's **160** downward movement (such as the opposite direction of the first direction D1) is controlled, so as to improve the safety of the actuation system **150**. Furthermore, the plurality of cushioning members **192** can also provide a cushioning function, so that occupants or passengers sitting on the occupant platform **160** could have a more comfortable experience.

(23) It is worth noting that the actuation system **150** is not limited to the aforementioned arrangement. According to another embodiment of the present invention, the actuation system **150** may also have a plurality of motion simulating apparatuses **100** arranged asymmetrically (not shown in FIGS.) to drive the occupant platform **160** to move and/or rotate.

(24) Advantages of the structures described herein include the ability to provide a motion simulating apparatus that is relatively simple in construction, and can be manufactured with reduced cost. Moreover, the motion simulating apparatus described can move smoothly and stably, and the height of the motion simulating apparatus can be reduced such that the installation space occupied by the motion simulating apparatus can also be reduced. Therefore, the motion simulating apparatus may be particularly suitable for use in environments of limited space. According to above arrangement, the motion simulating apparatus described herein is suitable for use with actuating systems to achieve the motion experiences of heave and roll.

(25) Realizations of the structures have been described only in the context of particular embodiments. These embodiments are meant to be illustrative and not limiting. Many variations, modifications, additions, and improvements are possible. Accordingly, plural instances may be provided for components described herein as a single instance. Structures and functionality presented as discrete components in the exemplary configurations may be implemented as a combined structure or component. These and other variations, modifications, additions, and improvements may fall within the scope of the claims that follow.

Claims

1. A motion simulating apparatus comprising: a base; a movable platform arranged above the base; a first cross arm assembly, disposed on a first side between the base and the movable platform, wherein the first cross arm assembly includes a first arm and a second arm that are cross-connected to each other; a second cross arm assembly, disposed on a second side between the base and the movable platform relative to the first side, wherein the second cross arm assembly includes a third arm and a fourth arm that are cross-connected to each other; an actuator, connected obliquely to an upper part of the first arm and the base or a lower part of the third arm, configured to drive the movable platform to move along a first direction or an opposite direction of the first direction; and a rod, horizontally connected with the upper part of the first arm and an upper part of the third arm, wherein the base, the movable platform, the first arm and the third arm substantially form a parallelogram when the actuator is actuating.
2. The motion simulating apparatus of claim 1, further comprising: a first roller assembly, disposed on the upper part of the first arm and a lower part of the second arm; and a second roller assembly, disposed on the upper part of the third arm and a lower part of the fourth arm, wherein the first roller assembly and the second roller assembly are configured to be moved along a second direction or an opposite direction of the second direction when the actuator is actuating, so as to elevate the movable platform.
3. The motion simulating apparatus of claim 2, wherein when the actuator is actuating, the movable platform and the rod are substantially parallel to the second direction.
4. The motion simulating apparatus of claim 1, wherein the first arm and the third arm are substantially parallel to each other and respectively connected to the base at two fixed ends, and the length of the rod is substantially equal to the distance between the two fixed ends.
5. The motion simulating apparatus of claim 1, wherein the first arm comprises a first sub-arm and a second sub-arm spaced apart along a third direction, and the first sub-arm and the second sub-arm are pivotally connected through a first pivot member; and wherein the second arm comprises a third sub-arm and a fourth sub-arm spaced apart along the third direction, and the third sub-arm and the fourth sub-arm are pivotally connected through the first pivot member.
6. The motion simulating apparatus of claim 1, wherein the third arm comprises a fifth sub-arm and a sixth sub-arm spaced apart along a third direction, and the fifth sub-arm and the sixth sub-arm are pivotally connected through a second pivot member; and wherein the fourth arm comprises a seventh sub-arm and a eighth sub-arm spaced apart along the third direction, and the seventh sub-arm and the eighth sub-arm are pivotally connected through the second pivot member.
7. The motion simulating apparatus of claim 5, wherein the first sub-arm and the third sub-arm are cross-connected to each other, and the second sub-arm and the fourth sub-arm are cross-connected to each other.
8. The motion simulating apparatus of claim 6, wherein the fifth sub-arm and the seventh sub-arm are cross-connected to each other, and the sixth sub-arm and the eighth sub-arm are cross-connected to each other.
9. The motion simulating apparatus of claim 2, further comprising: a first supporting member, disposed between the second arm and the base; and a second supporting member, disposed between the fourth arm and the base; wherein the first supporting member is configured to support the first roller assembly, and the second supporting member is configured to support the second roller assembly.
10. The motion simulating apparatus of claim 5, wherein the first direction, the second direction and the third direction are substantially orthogonal to each other.
11. The motion simulating apparatus of claim 6, wherein the first direction, the second direction and the third direction are substantially orthogonal to each other.
12. The motion simulating apparatus of claim 2, wherein the first cross arm assembly forms an X-shaped connection, the second cross arm assembly forms an X-shaped connection, and the first

cross arm assembly and the second cross arm assembly are arranged in a same direction.

13. An actuating system comprising: an occupant platform, adapted to carry one or more occupants; and a plurality of the motion simulating apparatus of claim 1, wherein the movable platform of the motion simulating apparatus is pivotally connected to the occupant platform through a plurality of joint assemblies.

14. The actuating system of claim 13, wherein the actuator of the motion simulating apparatus is operable to drive the occupant platform to move along the first direction or the opposite direction of the first direction.

15. The actuating system of claim 13, wherein the actuator of the motion simulating apparatus is operable to drive the occupant platform to rotate about a pivot axis of the second direction.

16. The actuating system of claim 13, further comprising: a plurality of slide assemblies, disposed between the occupant platform and the plurality of joint assemblies, wherein the occupant platform is substantially moved along the third direction or the opposite direction of the third direction relative to at least one of the plurality of joint assemblies.
